

MUHAMMAD ABDULLAH

+92301 8906387 abdullahbhatti6511@gmail.com linkedin.com/in/Abdullah github.com/abdullah

EDUCATION

FAST National University of Computer and Emerging Sciences

Bachelor of Science in Computer Science

Lahore, Pakistan

Aug. 2022 – Jun. 2026

Forman Christian College

FSc Pre-Engineering

Lahore, Pakistan

Aug. 2019 – May 2021

EXPERIENCE

Core Edge Solutions

Full Stack AI Intern

Jun. 2025 – Aug. 2025

Lahore, Pakistan

- Developed production-level web applications using React.js, FastAPI, Node.js, and PostgreSQL.
- Built an AI-powered querying platform for PDFs and databases using Retrieval-Augmented Generation (RAG) and semantic search.
- Engineered a multi-agent AI routing system that dynamically selected LLMs based on persona-driven context, reducing response latency by 35%.
- Developed features for an AI-driven accounting platform that automated invoice processing and email workflows using natural language commands.

PROJECTS

ZeroQ – Smart Retail Checkout System | Computer Vision

- Built a “Just Walk Out” retail checkout prototype using hybrid sensor fusion (Computer Vision, RFID, and Weight Sensors) to automate customer tracking and eliminate manual checkout.
- Developed a React Native mobile application with real-time virtual cart synchronization and automated payment workflows.
- Designed scalable backend services to support real-time user activity tracking and sensor communication under concurrent user load.

Dynamic Agentic Systems | FastAPI, LangGraph, React.js, Pinecone, PostgreSQL

- Built an AI-powered platform for intelligent querying across documents and databases using FastAPI, LangGraph, Pinecone, and PostgreSQL.
- Implemented dynamic AI persona routing with specialized assistants including Financial Analyst, Legal Advisor, and General Assistant using multi-agent workflows.
- Developed Retrieval-Augmented Generation (RAG) pipelines with semantic vector search, metadata tracing, and real-time query suggestions.
- Designed a modern three-panel React interface with document upload, conversational querying, metadata visualization, and query trace monitoring.

BookSphere – E-Library Platform | PostgreSQL, Express.js, React.js, Node.js

- Built a PERN stack e-library platform with advanced search, filtering, and personalized bookshelves.
- Developed secure RESTful APIs with JWT authentication and role-based access control.
- Designed PostgreSQL relational database schemas for scalable content and user management.

AI Workout Planner Web App | React.js, Node.js, Express.js, MongoDB

- Developed a full-stack workout planning and tracking platform with community-driven features.
- Designed scalable MongoDB schemas and built RESTful APIs for workout scheduling and progress tracking.
- Implemented reusable frontend components and responsive UI architecture using React.js.

TECHNICAL SKILLS

Languages & Frameworks: C++, Python, SQL, Node.js, FastAPI, Flask, React, Next.js, Express, MERN, Supabase

AI & Infrastructure: LangChain, LangGraph, OpenAI APIs, Pinecone, RAG, Multi-Agent Systems

Developer Tools: Git, GitHub, VS Code, PyCharm, Jupyter Notebook